深圳大学
SHENZHEN UNIVERSITY

模式识别
Pattern Recognition

——与计算机视觉
and Computer Vision

樊超

大数据系统计算技术国家工程实验室@深圳大学

E-mail：chaofan996@szu.edu.cn

特征提取与选择

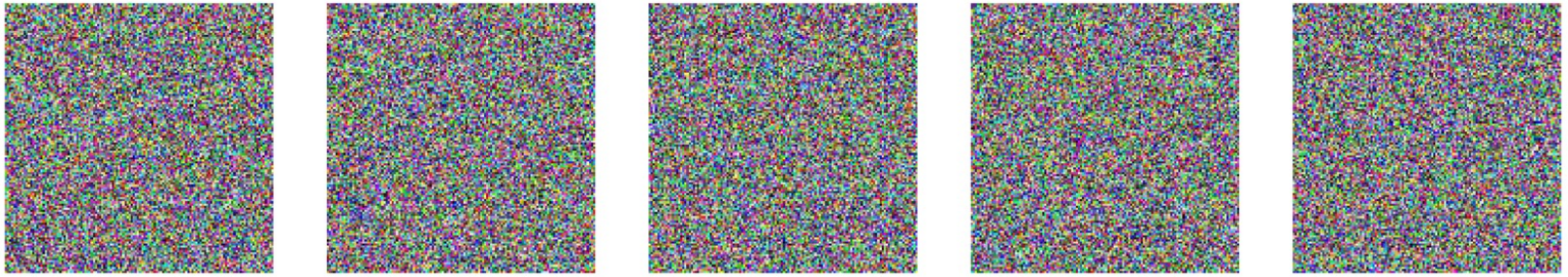
The **curse of dimensionality**

200×200像素的RGB图像

有10^{96329}种可能性！！！



特征提取与选择



自然图像仅占像素空间的极小部分——

随机采样200×200 个像素，能组成自然图像的概率几乎是0



特征提取与选择



逼真且平滑的图像变换（Image Transformation）——流形上的一条连续路径

流形假说（Manifold Hypothesis）认为：

数据密度集中在低维流形附近，

可以将高维 d 转化为低维 d_M，其中$d \ll d_M$

图片来源：Li, Xiaohui, et al. "Realistic Facial Age Transformation with 3D Uplifting." *Computer graphics forum*. Vol. 43. No. 4. 2024.

特征表示的重要性

Overview Data Code Models Discussion Leaderboard Rules

Leaderboard

Raw Data Refresh

Search leaderboard

Public Private

The private leaderboard is calculated with approximately 70% of the test data.
This competition has completed. This leaderboard reflects the final standings.

#	△	Team	Members	Score	Entries	Last Solution
1	—	Pierre Sermanet		0.98914	5	11y
2	▲ 4	orchid		0.98308	17	11y
3	—	Owen		0.98171	15	11y

KAGGLE · PLAYGROUND PREDICTION COMPETITION · 11 YEARS AGO

Late Submission

Dogs vs. Cats

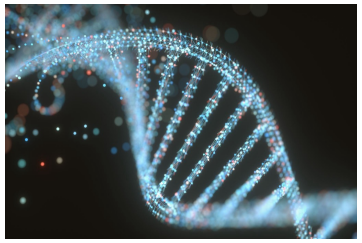
Create an algorithm to distinguish dogs from cats



判别性（Discriminative）

特征表示的重要性

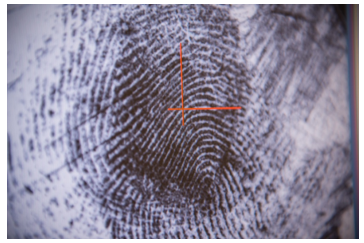
以人的识别特征为例，不同特征的适用性不同



（a）DNA



（b）人脸



（c）指纹



（d）虹膜



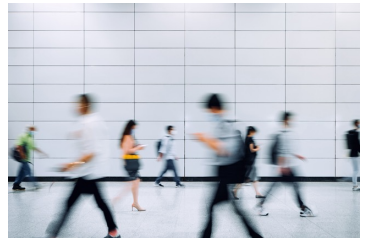
（e）掌纹



（f）签名

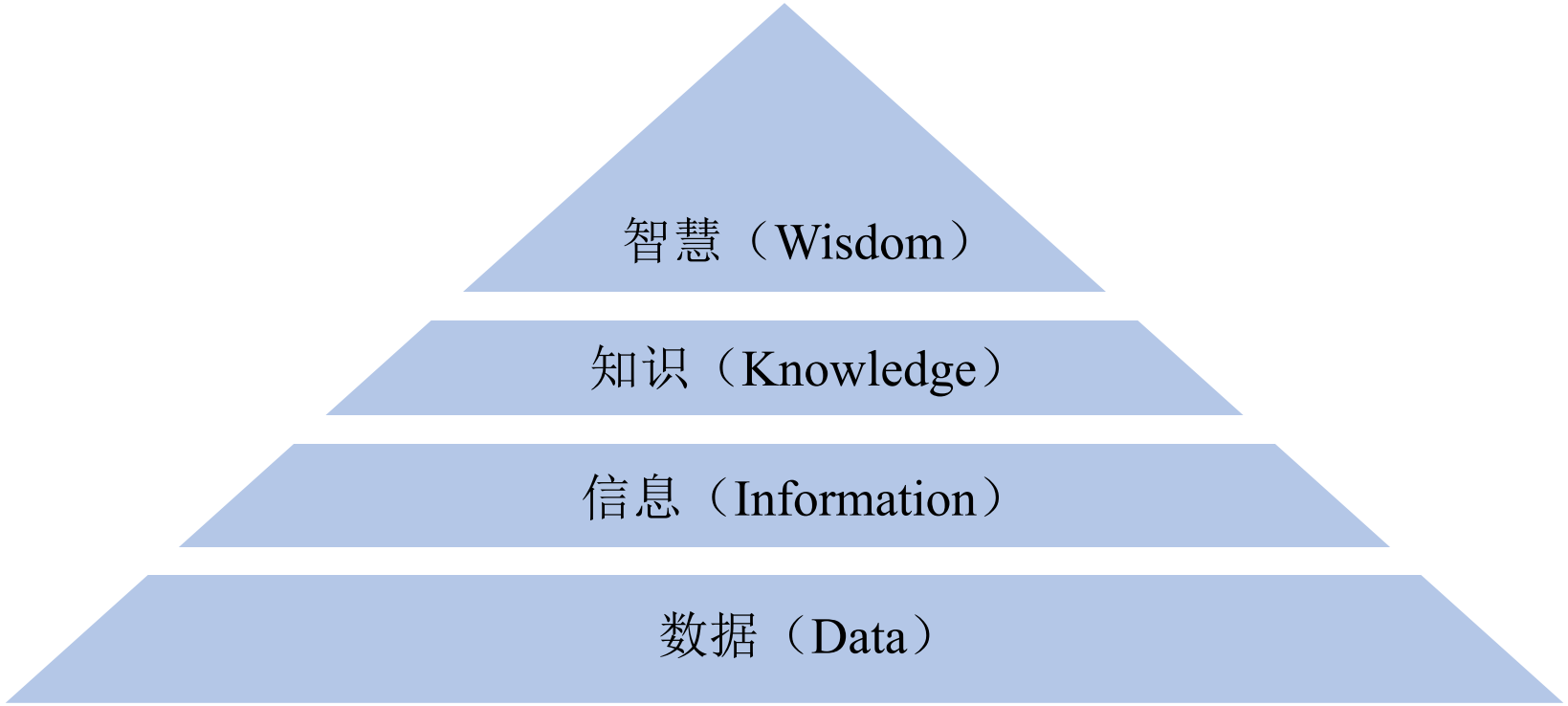


（g）声纹



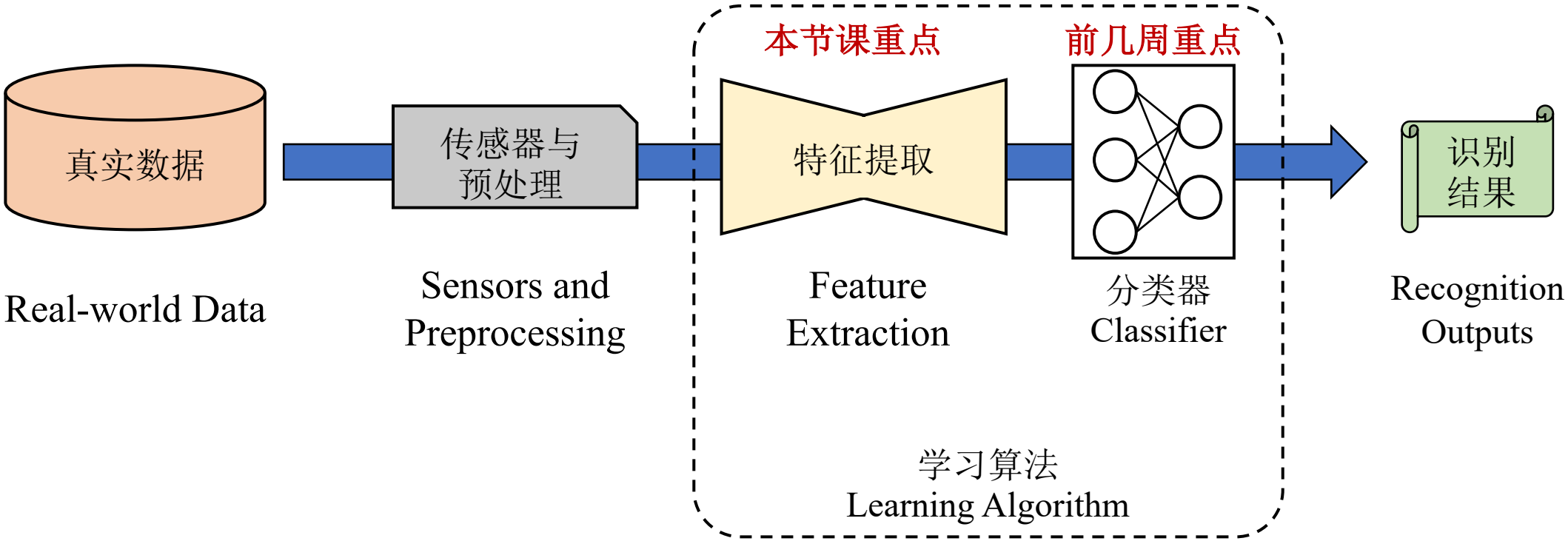
（h）步态

特征表示的重要性



特征的层级结构

特征表示的重要性


本节课重点
前几周重点
真实数据
传感器与
预处理
特征提取
识别
结果
Real-world Data
Sensors and
Preprocessing
Feature
Extraction
分类器
Classifier
Recognition
Outputs
学习算法
Learning Algorithm

自然图像中的挑战

view point variation



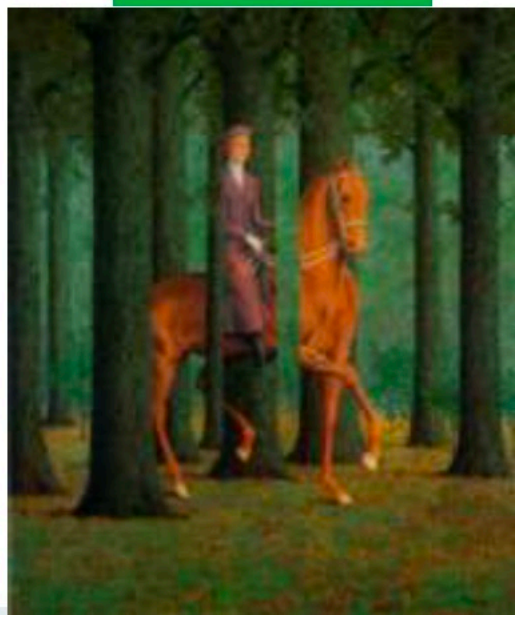
Michelangelo 1475-1564

illumination



自然图像中的挑战

occlusion



Magritte, 1957

scale



自然图像中的挑战

deformation



XU Beihong, 1957

background clutter



Klimt, 1957

经典图像特征：SIFT算子





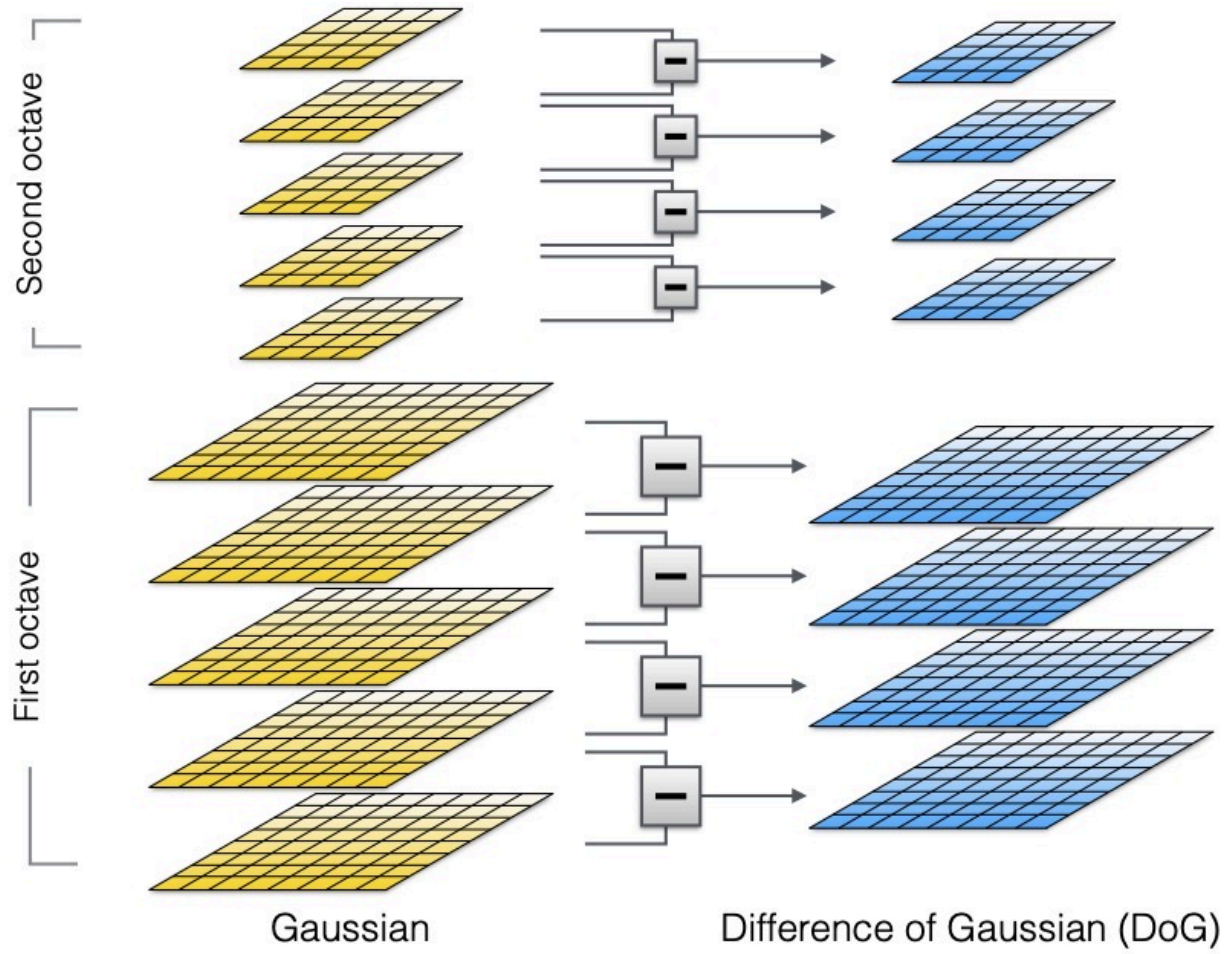
SIFT算子(Scale Invariant Feature Transform)

1. 多尺度极值检测（Multi-scale Extrema Detection）
2. 关键点定位（Keypoint Localization）
3. 方向分配（Orientation Assignment）
4. 关键点算子（Keypoint Descriptor）

参考文献：https://www.cs.cmu.edu/~16385/s17/Slides/7.3_SIFT_Detector_and_Descriptor.pdf

经典图像特征：SIFT算子

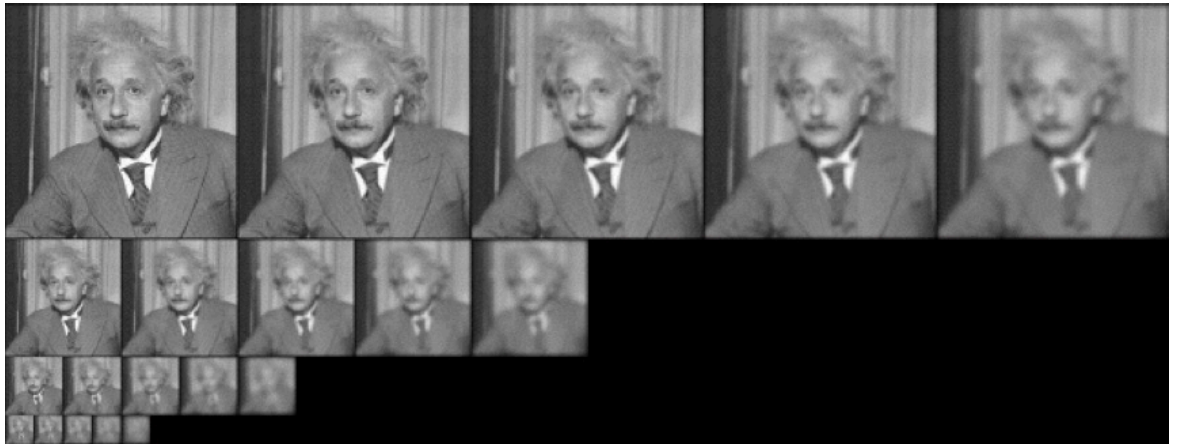
1. 多尺度极值检测（Multi-scale Extrema Detection）



经典图像特征：SIFT算子

1. 多尺度极值检测（Multi-scale Extrema Detection）

Gaussian

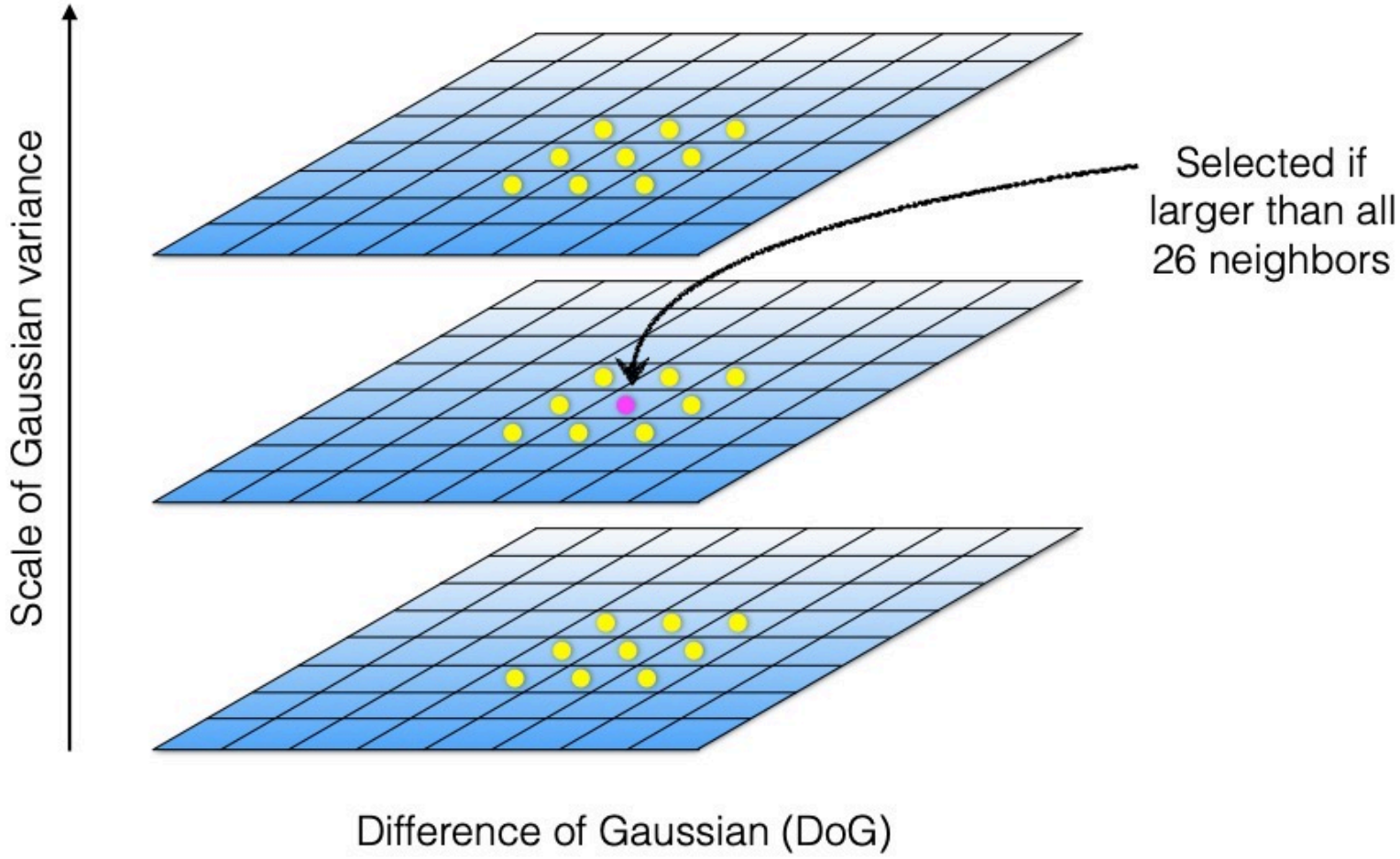


DoG, Difference of Gaussian



经典图像特征：SIFT算子

2. 关键点定位（Keypoint Localization）



经典图像特征：SIFT算子

3. 方向分配（Orientation Assignment）

For a keypoint, **L** is the **Gaussian-smoothed** image with the closest scale,

$$m(x,y) = \sqrt{(L(x+1,y) - L(x-1,y))^2 + (L(x,y+1) - L(x,y-1))^2}$$

x-derivative　　y-derivative

$$\theta(x,y) = \tan^{-1}((L(x,y+1) - L(x,y-1))/(L(x+1,y) - L(x-1,y)))$$

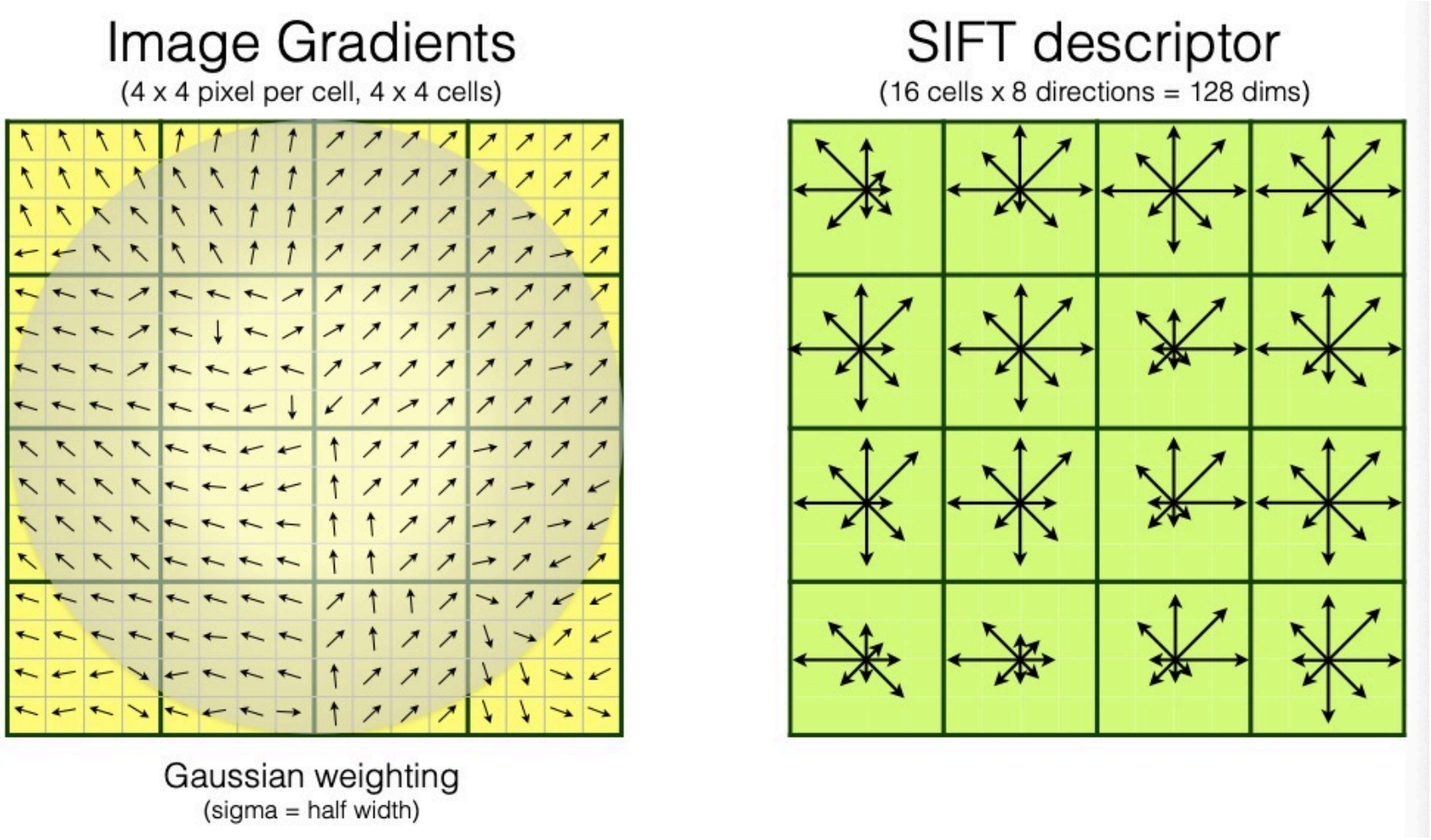
Detection process returns

$$\{x, y, \sigma, \theta\}$$

location　scale　orientation

经典图像特征：SIFT算子

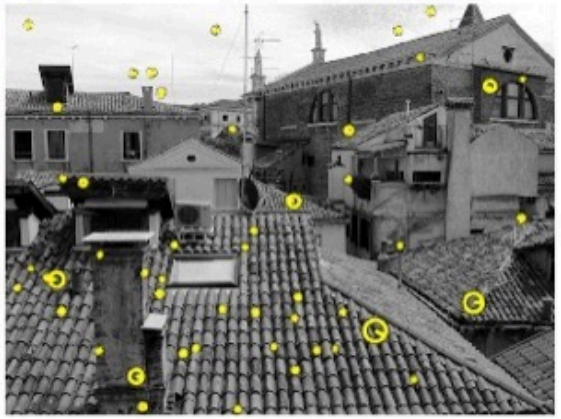
4. 关键点算子（Keypoint Descriptor）



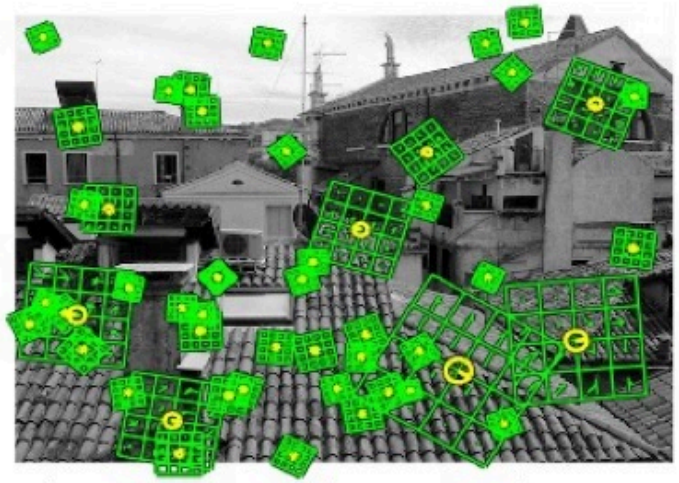
经典图像特征：SIFT算子



input



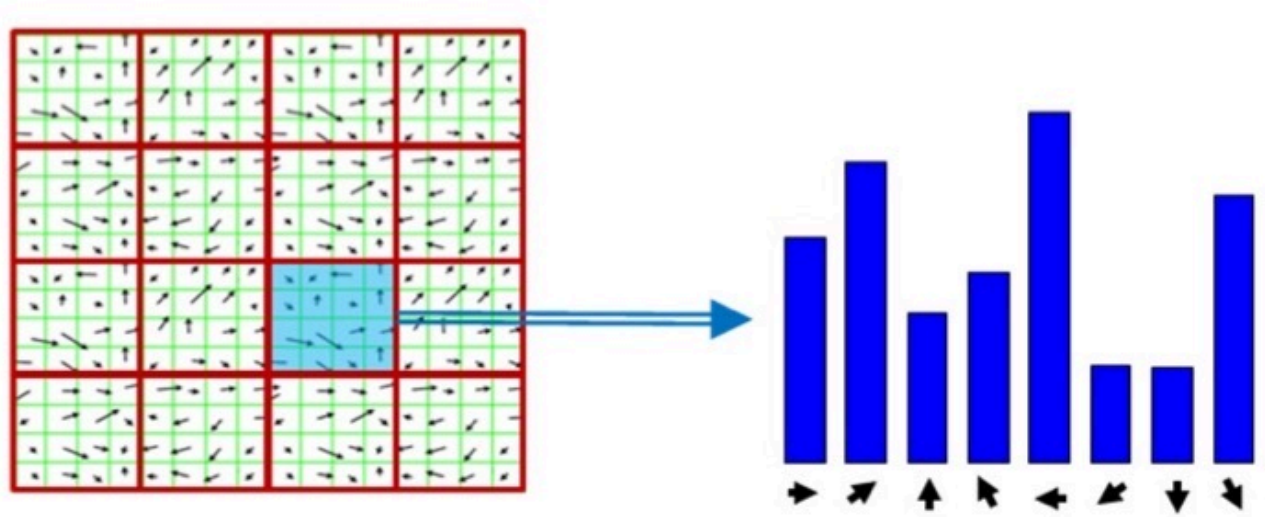
key points



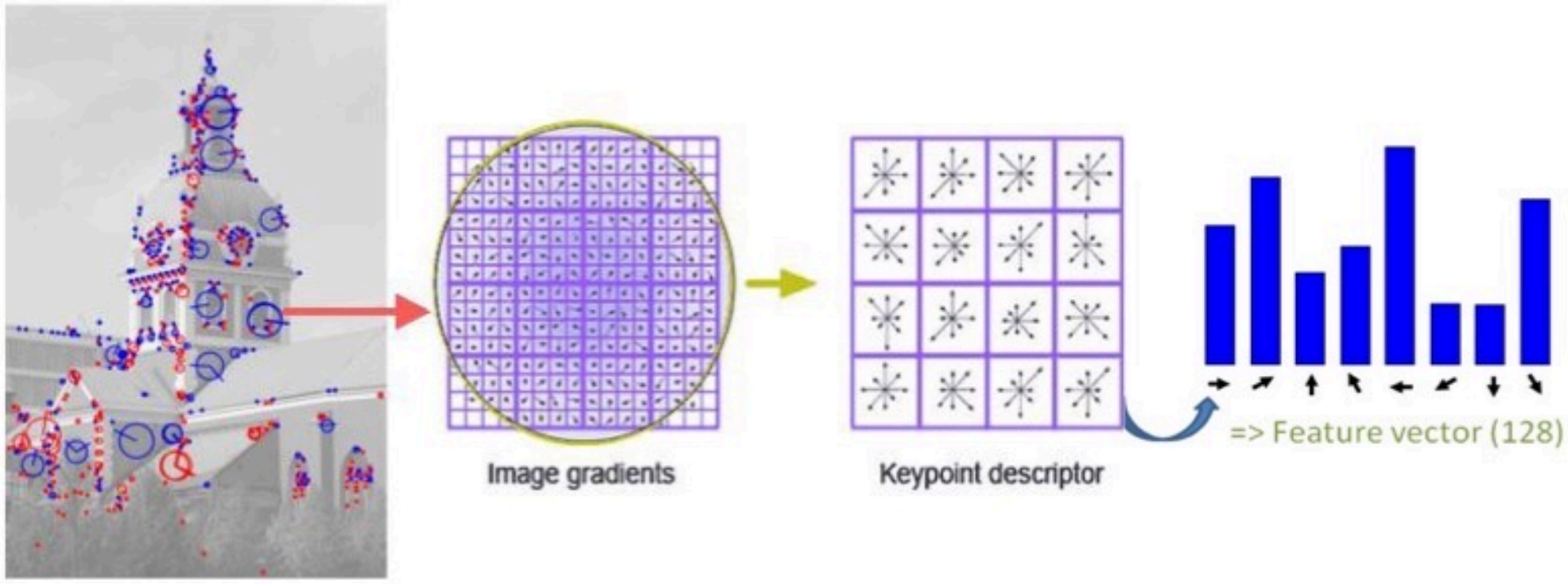
descriptor (regions)

经典图像特征：SIFT算子

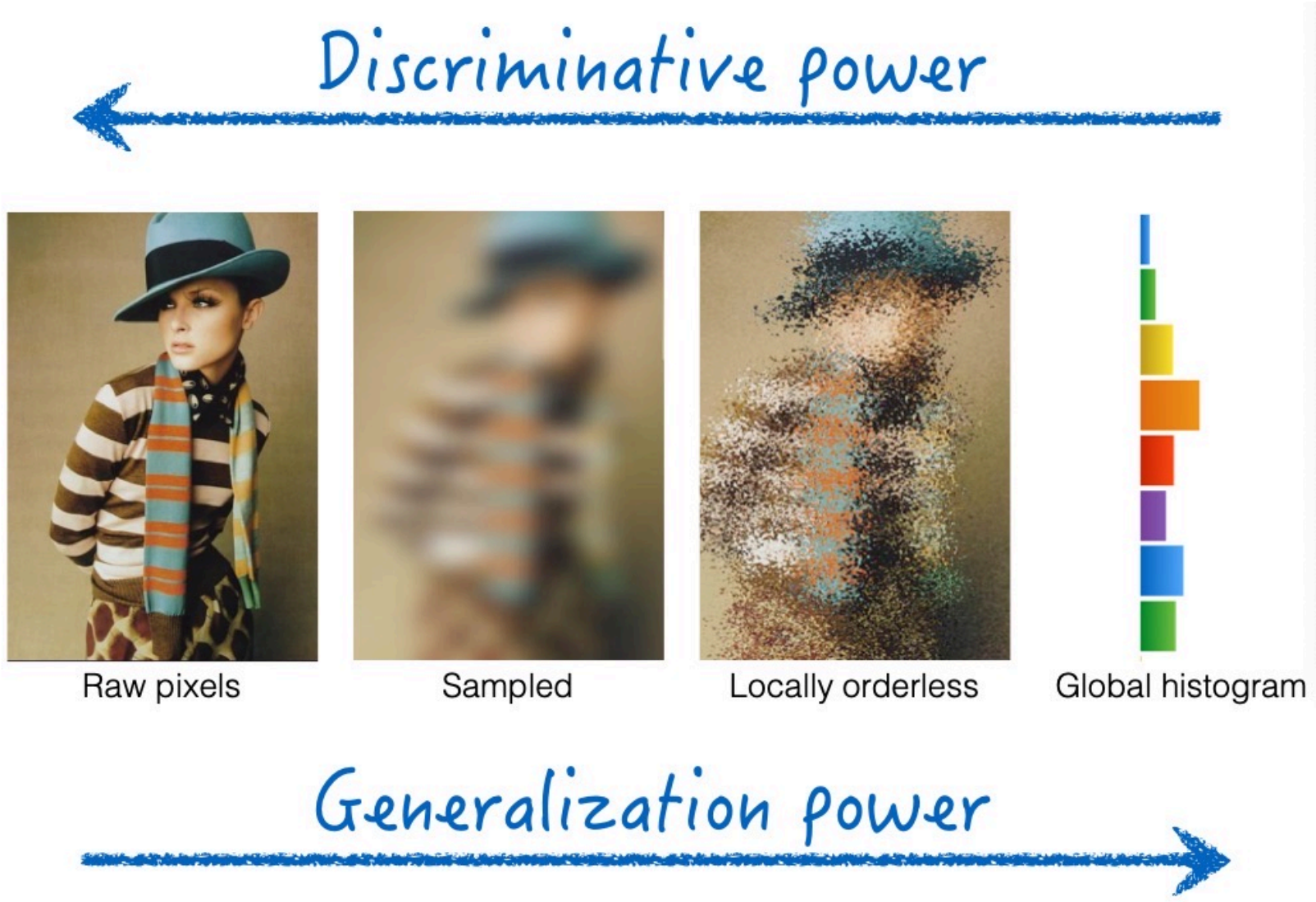
For each square, compute gradient direction histogram over 8 directions



经典图像特征：SIFT算子



经典图像特征：SIFT算子



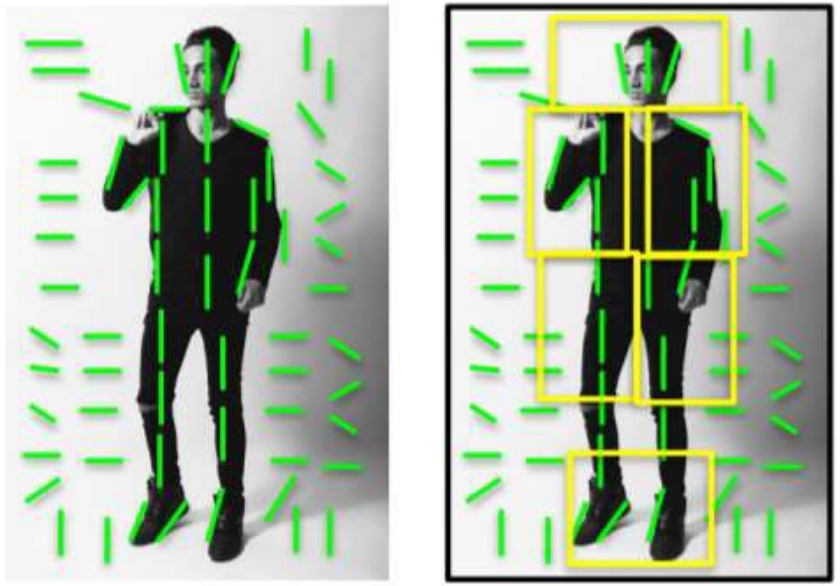
经典图像特征

还有很多经典的图像算子，如Haar

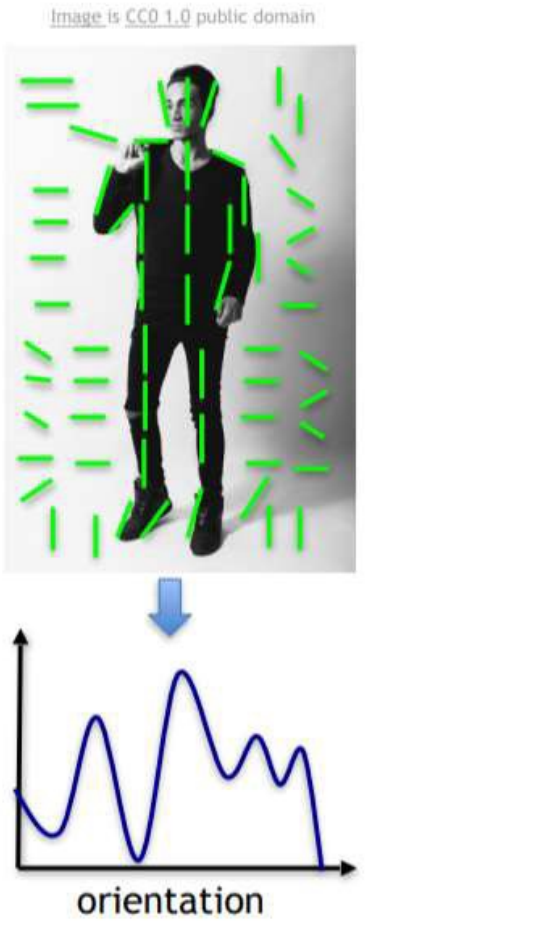


经典图像特征

还有很多经典的图像算子，如HOG



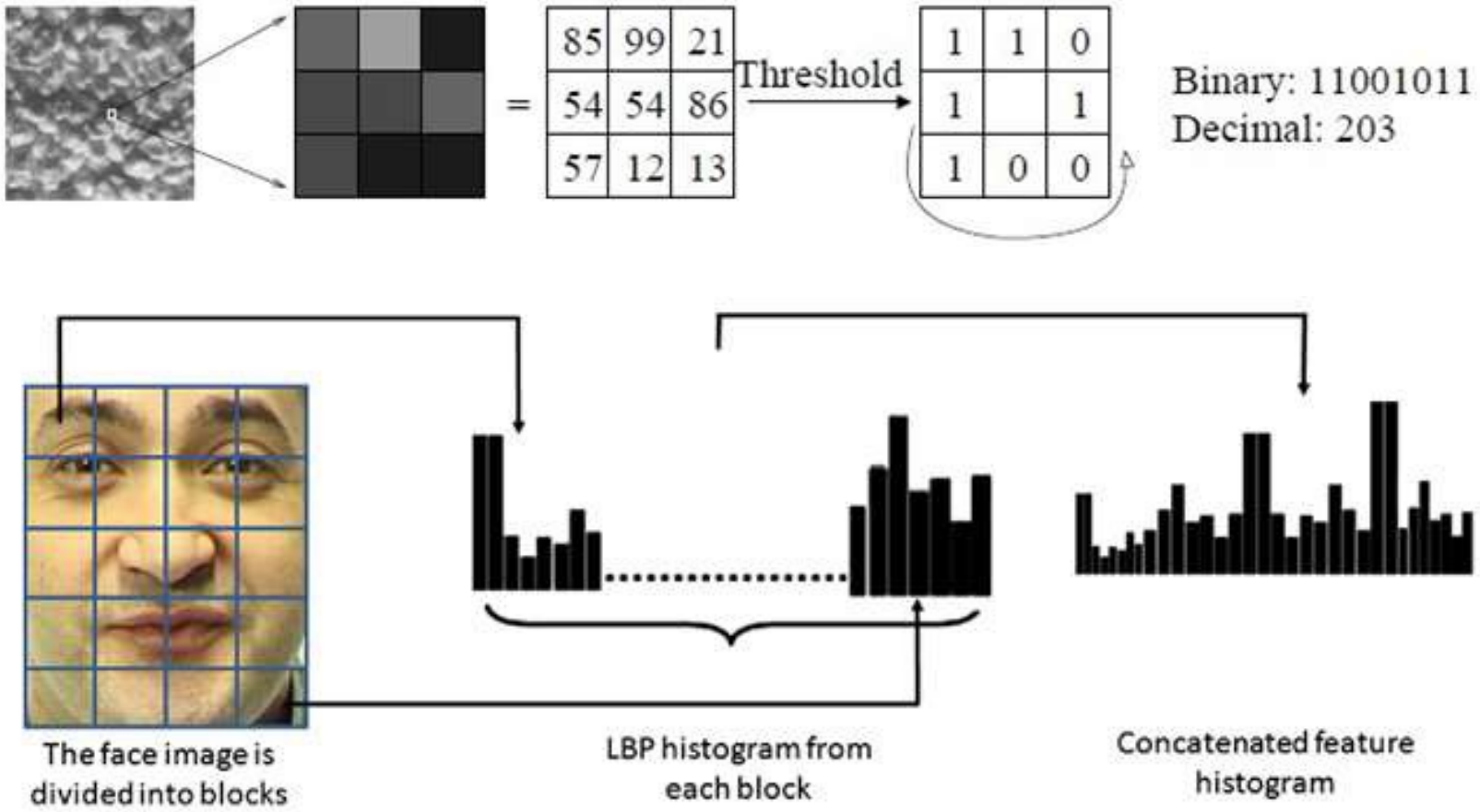
Deformable Part Model
Felzenswalb, McAllester, Ramanan, 2009



Histogram of Gradients (HoG)
Dalal & Triggs, 2005

经典图像特征

还有很多经典的图像算子，如LBP



特征的选择（维度削减）

Dimensionality reduction setup

Given n data points in d dimensions: $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^d$

$$\mathbf{X} = \begin{pmatrix} | & & | \\ \mathbf{x}_1 & \cdots\cdots & \mathbf{x}_n \\ | & & | \end{pmatrix} \in \mathbb{R}^{d \times n}$$

Want to reduce dimensionality from d to k

参考文献：http://people.eecs.berkeley.edu/~jordan/courses/pml/lectures/dimensionality/slides.pdf

主成分分析法（PCA）

Dimensionality reduction setup

Given n data points in d dimensions: $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^d$

$$\mathbf{X} = \begin{pmatrix} | & & | \\ \mathbf{x}_1 & \cdots\cdots & \mathbf{x}_n \\ | & & | \end{pmatrix} \in \mathbb{R}^{d \times n}$$

Want to reduce dimensionality from d to k

Choose k directions $\mathbf{u}_1, \ldots, \mathbf{u}_k$

$$\mathbf{U} = \begin{pmatrix} | & & | \\ \mathbf{u}_1 & \cdots & \mathbf{u}_k \\ | & & | \end{pmatrix} \in \mathbb{R}^{d \times k}$$

For each $\mathbf{u}_j$, compute "similarity" $z_j = \mathbf{u}_j^\top \mathbf{x}$

Project $\mathbf{x}$ down to $\mathbf{z} = (z_1, \ldots, z_k)^\top = \mathbf{U}^\top \mathbf{x}$

How to choose $\mathbf{U}$?

主成分分析法（PCA）

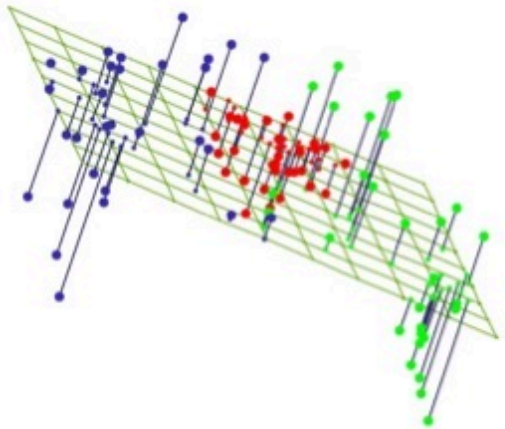
PCA objective 1: reconstruction error

$\mathbf{U}$ serves two functions:

- Encode: $\mathbf{z} = \mathbf{U}^\top \mathbf{x}, \quad z_j = \mathbf{u}_j^\top \mathbf{x}$
- Decode: $\tilde{\mathbf{x}} = \mathbf{U}\mathbf{z} = \sum_{j=1}^{k} z_j \mathbf{u}_j$

Want reconstruction error $\|\mathbf{x} - \tilde{\mathbf{x}}\|$ to be small

Objective: minimize total squared reconstruction error



$$\min_{\mathbf{U} \in \mathbb{R}^{d \times k}} \sum_{i=1}^{n} \|\mathbf{x}_i - \mathbf{U}\mathbf{U}^\top \mathbf{x}_i\|^2$$

主成分分析法（PCA）

PCA objective 2: projected variance

Empirical distribution: uniform over $\mathbf{x}_1, \ldots, \mathbf{x}_n$

Expectation (think sum over data points):

$$\hat{\mathbb{E}}[f(\mathbf{x})] = \frac{1}{n}\sum_{i=1}^{n} f(\mathbf{x}_i)$$

Variance (think sum of squares if centered):

$$\widehat{\text{var}}[f(\mathbf{x})] + (\hat{\mathbb{E}}[f(\mathbf{x})])^2 = \hat{\mathbb{E}}[f(\mathbf{x})^2] = \frac{1}{n}\sum_{i=1}^{n} f(\mathbf{x}_i)^2$$

Assume data is centered: $\hat{\mathbb{E}}[\mathbf{x}] = 0$ (what's $\hat{\mathbb{E}}[\mathbf{U}^\top \mathbf{x}]$?)

Objective: maximize variance of projected data

$$\max_{\mathbf{U} \in \mathbb{R}^{d \times k}, \mathbf{U}^\top \mathbf{U} = I} \hat{\mathbb{E}}[\|\mathbf{U}^\top \mathbf{x}\|^2]$$

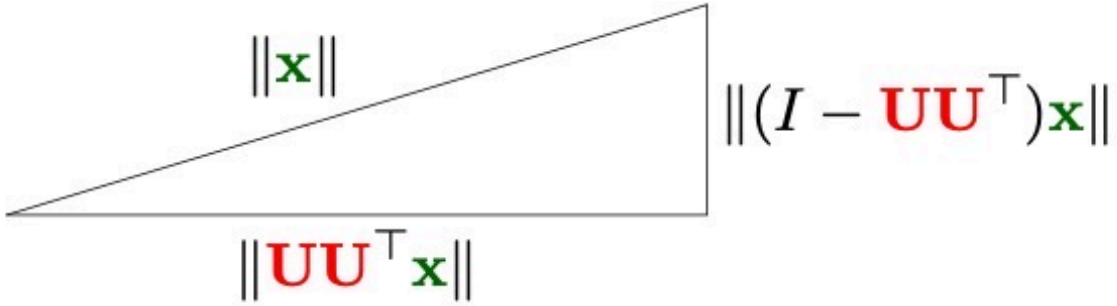
主成分分析法（PCA）

Equivalence in two objectives

Key intuition:

$$\underbrace{\text{variance of data}}_{\text{fixed}} = \underbrace{\text{captured variance}}_{\text{want large}} + \underbrace{\text{reconstruction error}}_{\text{want small}}$$

Pythagorean decomposition: $\mathbf{x} = \mathbf{U}\mathbf{U}^\top\mathbf{x} + (I - \mathbf{U}\mathbf{U}^\top)\mathbf{x}$



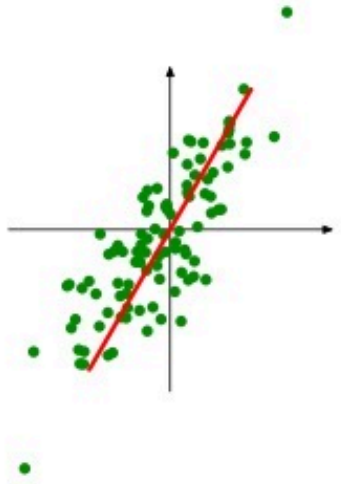
Take expectations; note rotation $\mathbf{U}$ doesn't affect length:

$$\hat{\mathbb{E}}[\|\mathbf{x}\|^2] = \hat{\mathbb{E}}[\|\mathbf{U}^\top\mathbf{x}\|^2] + \hat{\mathbb{E}}[\|\mathbf{x} - \mathbf{U}\mathbf{U}^\top\mathbf{x}\|^2]$$

Minimize reconstruction error $\leftrightarrow$ Maximize captured variance

主成分分析法（PCA）

Finding one principal component



Input data:

$$\mathbf{X} = \begin{pmatrix} | & & | \\ \mathbf{x}_1 & \cdots & \mathbf{x}_n \\ | & & | \end{pmatrix}$$

Objective: maximize variance of projected data

$$= \max_{\|\mathbf{u}\|=1} \hat{\mathbb{E}}[(\mathbf{u}^\top \mathbf{x})^2]$$

$$= \max_{\|\mathbf{u}\|=1} \frac{1}{n} \sum_{i=1}^{n} (\mathbf{u}^\top \mathbf{x}_i)^2$$

$$= \max_{\|\mathbf{u}\|=1} \frac{1}{n} \|\mathbf{u}^\top \mathbf{X}\|^2$$

$$= \max_{\|\mathbf{u}\|=1} \mathbf{u}^\top \left(\frac{1}{n}\mathbf{X}\mathbf{X}^\top\right) \mathbf{u}$$

$$= \text{largest eigenvalue of } C \stackrel{\text{def}}{=} \frac{1}{n}\mathbf{X}\mathbf{X}^\top$$

主成分分析法（PCA）

Computing PCA

Method 1: eigendecomposition

$\mathbf{U}$ are eigenvectors of covariance matrix $C = \frac{1}{n}\mathbf{X}\mathbf{X}^\top$

Computing C already takes $O(nd^2)$ time (very expensive)

Method 2: singular value decomposition (SVD)

Find $\mathbf{X} = \mathbf{U}_{d\times d}\Sigma_{d\times n}\mathbf{V}^\top_{n\times n}$

where $\mathbf{U}^\top\mathbf{U} = I_{d\times d}$, $\mathbf{V}^\top\mathbf{V} = I_{n\times n}$, Σ is diagonal

Computing top k singular vectors takes only $O(ndk)$

Relationship between eigendecomposition and SVD:

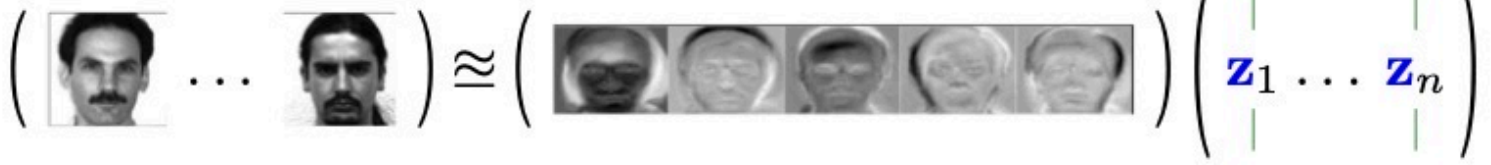
Left singular vectors are principal components ($C = \mathbf{U}\Sigma^2\mathbf{U}^\top$)

主成分分析法（PCA）：应用举例

Eigen-faces [Turk and Pentland, 1991]

- $d =$ number of pixels
- Each $\mathbf{x}_i \in \mathbb{R}^d$ is a face image
- $\mathbf{x}_{ji} =$ intensity of the j-th pixel in image i

$\mathbf{X}_{d\times n} \approx \mathbf{U}_{d\times k} \mathbf{Z}_{k\times n}$



Idea: $\mathbf{z}_i$ more "meaningful" representation of i-th face than $\mathbf{x}_i$

Can use $\mathbf{z}_i$ for nearest-neighbor classification

Much faster: $O(dk + nk)$ time instead of $O(dn)$ when $n, d \gg k$

主成分分析法（PCA）：应用举例

Latent Semantic Analysis [Deerwater, 1990]

- $d =$ number of words in the vocabulary
- Each $\mathbf{x}_i \in \mathbb{R}^d$ is a vector of word counts
- $\mathbf{x}_{ji} =$ frequency of word j in document i

$$\underset{\mathbf{X}_{d\times n}}{\begin{pmatrix} \text{stocks: } 2 & \cdots & 0 \\ \text{chairman: } 4 & \cdots & 1 \\ \text{the: } 8 & \cdots & 7 \\ \cdots & \vdots & \vdots \\ \text{wins: } 0 & \cdots & 2 \\ \text{game: } 1 & \cdots & 3 \end{pmatrix}} \approx \underset{\mathbf{U}_{d\times k}}{\begin{pmatrix} 0.4 & \cdots & -0.001 \\ 0.8 & \cdots & 0.03 \\ 0.01 & \cdots & 0.04 \\ \vdots & \cdots & \vdots \\ 0.002 & \cdots & 2.3 \\ 0.003 & \cdots & 1.9 \end{pmatrix}} \underset{\mathbf{Z}_{k\times n}}{\begin{pmatrix} | & & | \\ \mathbf{z}_1 & \cdots & \mathbf{z}_n \\ | & & | \end{pmatrix}}$$

How to measure similarity between two documents?

$\mathbf{z}_1^\top \mathbf{z}_2$ is probably better than $\mathbf{x}_1^\top \mathbf{x}_2$

Applications: information retrieval

维度削减

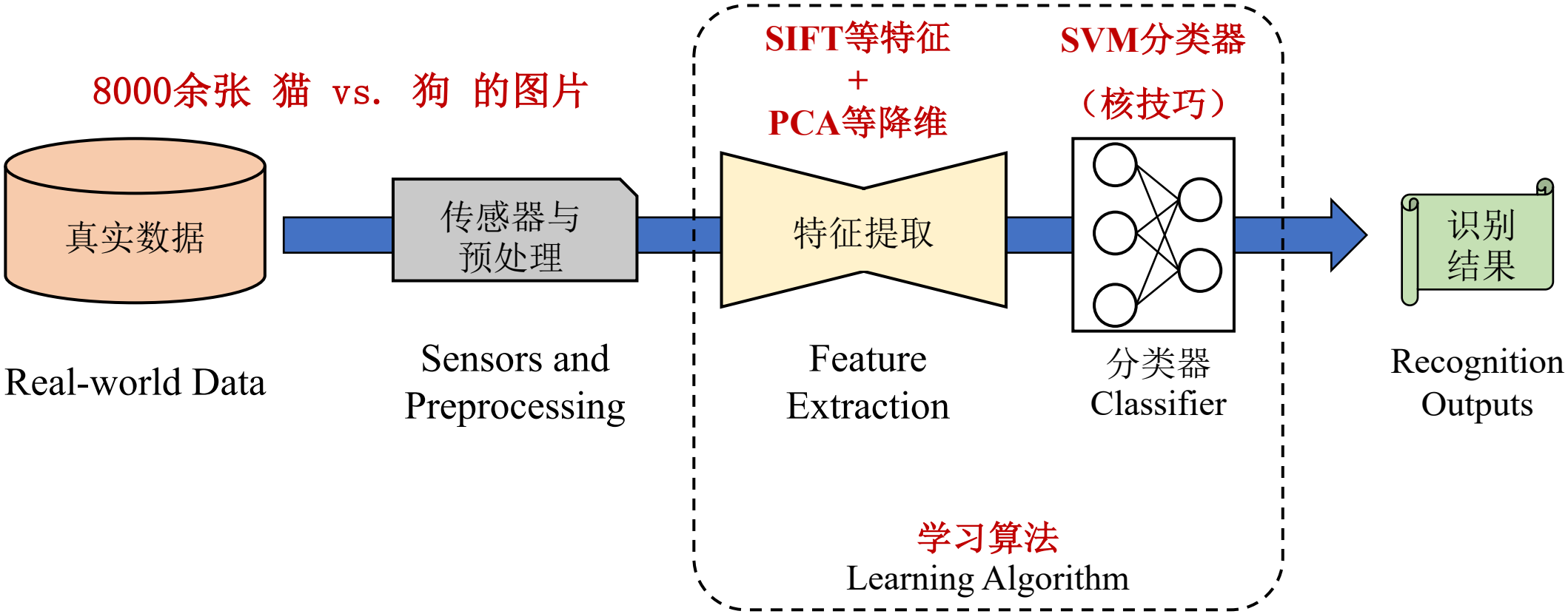
Motivation and context

Why do dimensionality reduction?

- Computational: compress data ⇒ time/space efficiency
- Statistical: fewer dimensions ⇒ better generalization
- Visualization: understand structure of data
- Anomaly detection: describe normal data, detect outliers

后续会介绍更多降维算法与应用！！！

第7周：第2次随堂测试


8000余张 猫 vs. 狗 的图片
SIFT等特征
+
PCA等降维
SVM分类器
（核技巧）
真实数据
传感器与
预处理
特征提取
识别
结果
Real-world Data
Sensors and
Preprocessing
Feature
Extraction
分类器
Classifier
Recognition
Outputs
学习算法
Learning Algorithm